The Probability of Magic The Gathering

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Handmade Infographic

Full Logo Url, Small Logo Url

60
Deck Size

24
Lands

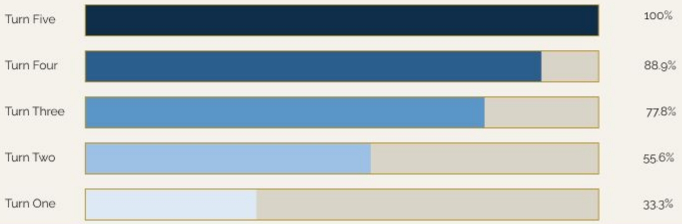
MAGIC The Gathering®

7
Cards Drawn

4
Copies per Deck

Mana Curve: Spells Castable by Turn

Castable Spells by Turn



Opening Hand: Probability of a Good Opener

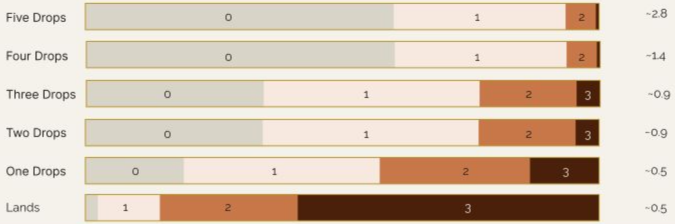
56.5%
Ideal opener: 2–3 lands + ≥1 castable spell

80.9%
P(≥1 one-drop)

65.4%
P(≥1 two-drop)

78.9%
P(castable turn 1)

93.0%
P(castable turn 2)



Why Four Copies: Copy Count Probabilities

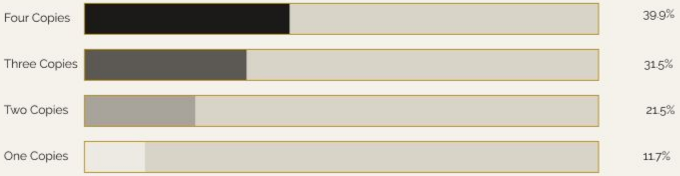
39.9%
At least 1 copy in opening hand

60.1%
No copies

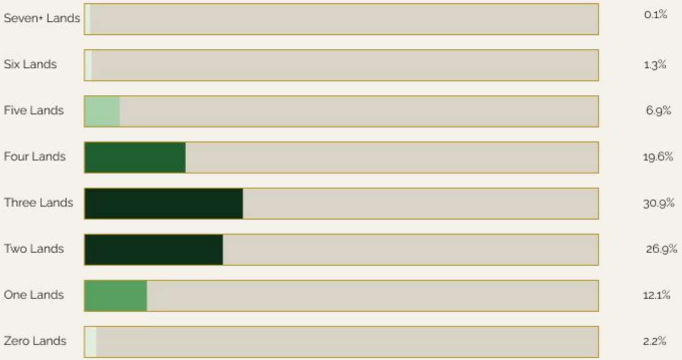
33.6%
Exactly 1

6.3%
2 or more

Opening Hand Odds by Copy Count



Land Distribution: Land Count in Opening



Data: The data used in this infographic was not collected empirically but derived using hypergeometric distribution: using this script

Statistics: The statistics i used in this infographic are descriptive and probabilistic. Describing the theoretical distribution of known outcomes in a fixed system. With a 60 card deck with a fully defined population, allows me to calculate exact probabilities.

Accuracy of conclusion:Mathematically sound given stated assumptions. I chose to use a hypergeometric distribution model here because it describes drawing from a finite population without replacement, which happens when you draw an opening hand from a shuffled deck, in a Trading Card Game. While a binomial distribution assumes each draw is independent, while both would produce the same expected values, the Hypergeometric gives us a more accurate (smaller) variance by taking into account a population correction factor each time a card is drawn.

Illuminating information:

- Mana Curve: The mana curve section illustrates the evolution of a deck's playability over the first five turns. Notably, even with 24 lands, only 33.3% of opening hands can cast a spell on turn one, increasing to 100% by turn five. This progression highlights the impact of a top-heavy curve, which becomes apparent through this analysis rather than through raw card counts alone.
- Opening Hand: The opening hand section quantifies the criteria for retaining an initial hand. A 56.5% probability of drawing a strong opener indicates that even optimally constructed decks fail to produce an ideal starting hand nearly half the time. This finding underscores the rationale for the mulligan rule and emphasizes the significance of the first seven cards in determining game outcomes.
- Why Four Copies: This section demonstrates that the four-copy maximum in competitive Magic is a deliberate design choice. Increasing from one copy (11.7%) to four copies (39.9%) nearly quadruples the probability of drawing a key card in the opening hand. This result indicates that consistency is primarily a function of intentional deck construction rather than chance.
- Land Distribution: The land distribution section quantifies the risks of mana screw and mana flood. Although lands comprise 40% of the deck, there remains a 2.2% probability of drawing zero lands and an 8.3% probability of drawing five or more. These small but significant probabilities justify careful consideration of land counts and the use of the mulligan rule.

Caution: The primary caution is that this infographic assumes familiarity with Magic: The Gathering. Terms like "mana curve" and "good opener" are never defined, so the statistics lose significance without prior game knowledge. The probabilities represent a theoretical baseline, assuming a perfectly random draw with no mulligans, tutors, mana rocks, or card draw effects common in real gameplay. Finally, the 56.5% "good opener" figure is more subjective than it appears. It reflects one specific definition of a keepable hand that not all players or deck archetypes would agree on.